# Customer Behavior Analysis Report

## Project Overview

This project analyzes customer shopping behavior using Python (Pandas), PostgreSQL, and Power BI. The goal is to extract meaningful insights from customer data and present them through an interactive dashboard.

## Tools and Technologies

- Python (Pandas) for data cleaning and feature engineering
- PostgreSQL for data storage and SQL-based analysis
- Power BI for dashboard visualization

## Data Preparation

The dataset was first processed in Python. Missing values in the review rating column were handled by filling them with the median value based on product category. Column names were standardized to lowercase with underscores for easier handling. New features were created, including age_group using quantile-based binning and purchase_frequency_days by mapping purchase frequency categories to numeric values. The redundant column promo_code_used was removed.

## Database Integration

The cleaned dataset was uploaded to PostgreSQL using SQLAlchemy. The data was stored in a table named customer. SQL queries were then used to perform analysis directly on the database.

## Analysis Questions and Insights

**Q1.** Total revenue generated by male vs female customers

This analysis calculates total purchase amount grouped by gender. It helps identify which gender contributes more to overall revenue and supports targeted marketing strategies.

| | gender text | total_revenue numeric |
|---|---|---|
| 1 | Female | 75191 |
| 2 | Male | 157890 |

**Q2.** Customers who used a discount but still spent more than the average purchase amount

This identifies high-value customers who used discounts but still spent above the average purchase amount. These customers are important for retention and premium targeting.

**Q3.** Top 5 products with the highest average review rating

This analysis finds the most highly rated products based on average review scores. These products can be prioritized in promotions and recommendations.

| item_purchased text | avg_rating double precision |
|---|---|
| Gloves | 3.8614285714285725 |
| Sandals | 3.8443750000000003 |
| Boots | 3.8187500000000005 |
| Hat | 3.8012987012987005 |
| Skirt | 3.784810126582278 |

**Q4.** Comparison of average purchase amount between shipping types

This compares customer spending behavior based on shipping type. It helps determine whether faster or premium shipping options influence purchase value.

| shipping_type text | avg_purchase numeric |
|---|---|
| Express | 60.4752321981424149 |
| Next Day Air | 58.6311728395061728 |
| Store Pickup | 59.8938461538461538 |
| 2-Day Shipping | 60.7336523125996810 |
| Free Shipping | 60.4103703703703704 |
| Standard | 58.4602446483180428 |

**Q5.** Subscriber vs non-subscriber spending behavior

This compares average spending and total revenue between subscribed and non-subscribed customers. The insight helps evaluate the effectiveness of subscription programs.

| subscription_status text | avg_spend numeric | total_revenue numeric |
|---|---|---|
| No | 59.8651211801896733 | 170436 |
| Yes | 59.4919278252611586 | 62645 |

**Q6.** Products with the highest percentage of discounted purchases

This identifies products most frequently bought with discounts. It provides insights into pricing sensitivity and promotional effectiveness.

| item_purchased text | discount_percentage numeric |
|---|---|
| Hat | 50.0000000000000000 |
| Sneakers | 49.6551724137931034 |
| Coat | 49.0683229813664596 |
| Sweater | 48.1707317073170732 |
| Pants | 47.3684210526315789 |

**Q7.** Customer segmentation based on purchase history

Customers are grouped into New, Returning, and Loyal segments based on the number of previous purchases. This segmentation helps understand customer lifecycle and retention strategies.

| customer_segment text | customer_count bigint |
|---|---|
| Loyal | 3476 |
| Returning | 424 |

**Q8.** Top 3 most purchased products within each category

This identifies the most popular products in each category using ranking. It helps in inventory management and category-level marketing decisions.

| category text | item_purchased text | purchase_count bigint |
|---|---|---|
| Accessori... | Jewelry | 171 |
| Accessori... | Sunglasses | 161 |
| Accessori... | Belt | 161 |
| Clothing | Blouse | 171 |
| Clothing | Pants | 171 |
| Clothing | Shirt | 169 |
| Footwear | Sandals | 160 |
| Footwear | Shoes | 150 |
| Footwear | Sneakers | 145 |
| Outerwear | Jacket | 163 |
| Outerwear | Coat | 161 |

**Q9.** Likelihood of repeat buyers being subscribers

This analysis checks whether customers with more than five previous purchases are more likely to subscribe. It helps understand the relationship between loyalty and subscription.

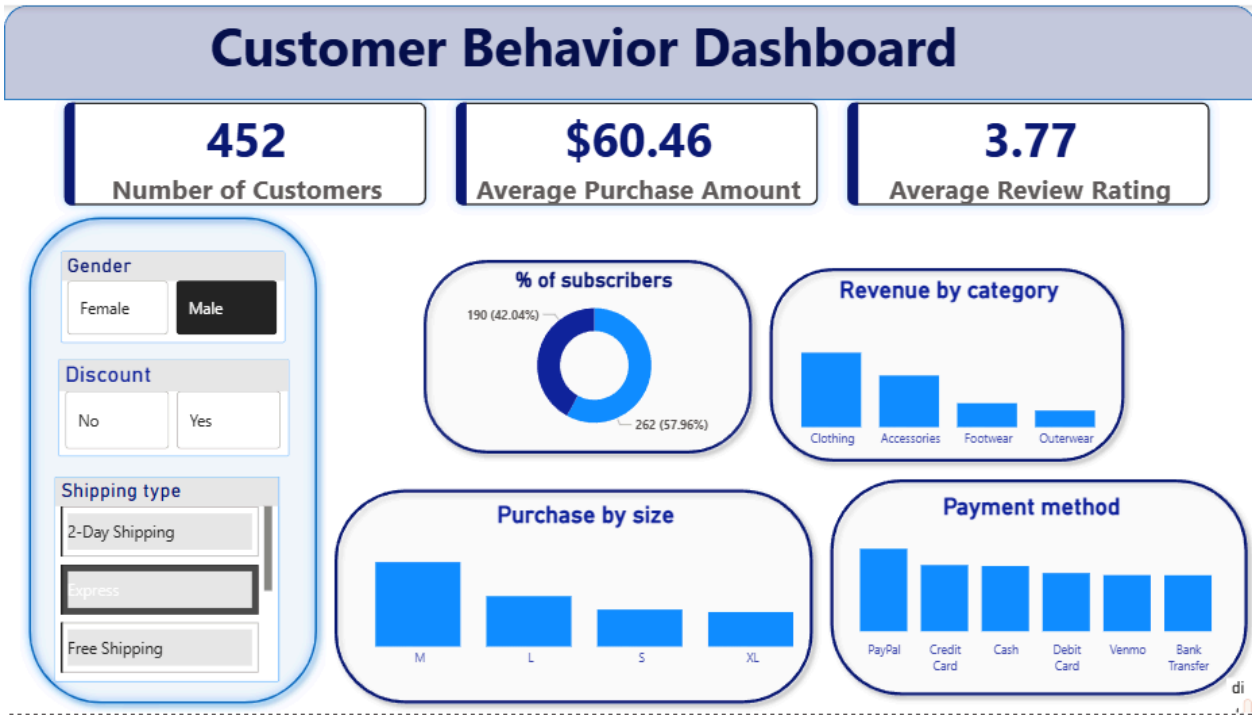
| subscription_status text | customer_count bigint | percentage numeric |
|---|---|---|
| No | 2518 | 72.4395857307249712 |
| Yes | 958 | 27.5604142692750288 |

**Q10.** Revenue contribution of each age group

This groups customers into age categories and calculates total revenue per group. It helps identify which age segment contributes the most to revenue.

| age_group text | total_revenue numeric |
|---|---|
| Young Adult | 62143 |
| Middle-aged | 59197 |
| Adult | 55978 |
| Senior | 55763 |

## Dashboard Overview




The dashboard provides a visual summary of the analysis. It includes key metrics such as total number of customers, average purchase amount, and average review rating. It also shows

revenue distribution by category, purchase patterns by size, and preferred payment methods. Interactive filters for gender, discount usage, and shipping type allow dynamic exploration of the data.

## Conclusion

The analysis shows that younger adult customers contribute the highest revenue. Subscription-based customers tend to spend more than non-subscribers. Discounts significantly influence purchasing behavior, especially for certain products. Clothing categories dominate overall sales. The project demonstrates how data-driven insights can support business decisions.

## Future Improvements

Future work could include predictive modeling for customer behavior, real-time dashboard integration, and advanced recommendation systems based on purchase patterns.